Improving Melanoma Classification Across Diverse Skin Tones Through Skin-Tone-Aware Image Augmentation

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Abstract—Automated melanoma classifiers can match dermatologist-level accuracy on curated benchmarks, yet they frequently underperform on darker skin tones because public dermatology datasets lack skin-tone diversity. This work investigates whether skin-tone-aware photometric image augmentation can improve classification robustness and fairness without designing a new architecture. An EfficientNet-B0 classifier, pretrained on ImageNet and fine-tuned on HAM10000, is evaluated as a cross-dataset generalization test on the Diverse Dermatology Images (DDI) set, which provides Fitzpatrick skin-tone labels. Two models are compared: a baseline trained with preprocessing only, and an augmented model trained with brightness, contrast, saturation, hue, flip, and rotation jitter. Grad-CAM overlays provide explainability. On DDI, augmentation improved malignant detection (F1 0.046 → 0.198, ROC-AUC 0.554 → 0.615) and narrowed the maximum–minimum per-group accuracy spread (0.077 → 0.049), but it did not reduce the Light-versus-Dark accuracy gap. The results suggest photometric augmentation is a weak proxy for genuinely diverse training data and that fairness gains most likely require representative images rather than synthetic color perturbation.

Index Terms—melanoma classification, skin-tone fairness, data augmentation, EfficientNet, Grad-CAM, computational imaging, medical image analysis.

I. INTRODUCTION

A. Problem and Motivation

Melanoma is one of the most severe forms of skin cancer, and early diagnosis significantly improves patient outcomes [1]. Advances in computational imaging and deep learning have enabled automated classification of skin lesions from clinical and dermoscopic images, and convolutional neural networks (CNNs) have become the dominant approach, demonstrating diagnostic performance comparable to dermatologists in some studies [1].

Despite these advances, existing skin-lesion classifiers often fail to generalize across diverse patient populations. A major limitation is the lack of skin-tone diversity in publicly available dermatology datasets [2]. Darker skin tones are frequently underrepresented, making it difficult to evaluate whether models perform consistently across all populations [2]. As a result, many systems exhibit reduced accuracy on skin-of-color (SOC) populations [2], [3]. Because patients with skin of color are already diagnosed at later stages, these disparities carry real clinical consequences, motivating the need for more equitable dermatology AI.

Recent work shows that state-of-the-art dermatology AI models perform worse on diverse clinical image sets containing darker skin tones, and that fine-tuning with more representative data reduces this gap [4]. This suggests improving dataset diversity may be as important as improving the model itself. Rather than developing a new classifier, this project investigates whether **skin-tone-aware image augmentation** — generating realistic variation in skin pigmentation and imaging conditions while preserving lesion characteristics — can improve melanoma classification across skin-tone subgroups.

B. Contributions

1. A reproducible cross-dataset pipeline that fine-tunes EfficientNet-B0 on HAM10000 and evaluates on the skin-tone-labeled DDI benchmark.
2. A controlled comparison of a baseline versus a photometrically augmented model on both overall performance and per-skin-tone fairness.
3. Grad-CAM–based explainability to verify the model attends to lesions, and a web application for interactive inference and results visualization.

II. RELATED WORK

CNNs for skin-lesion classification. Deep CNNs trained on dermoscopic datasets have achieved dermatologist-comparable accuracy for melanoma recognition [1]. The HAM10000 dataset [6] has become a standard training corpus of multi-source dermatoscopic images with biopsy-confirmed malignancies.

Dataset bias and skin-tone diversity. Reviews of skin-lesion datasets show pervasive under-representation of darker skin tones, limiting fairness evaluation [2]. Broader analyses of AI in dermatology for skin of color highlight both the promise and the persistent challenges of equitable performance [3].

Fairness evaluation by skin tone. The Diverse Dermatology Images (DDI) benchmark [4], [5] was curated specifically to expose these disparities. It is biopsy-confirmed and matched across Fitzpatrick skin types, and demonstrates that leading models — including one trained on HAM10000 — lose substantial ROC-AUC on Fitzpatrick V–VI skin, while fine-tuning on diverse data closes the gap [4].

Explainability. Grad-CAM produces class-discriminative localization maps from a trained classifier without extra labels, and is widely used to confirm that medical-imaging models focus on clinically relevant regions rather than artifacts.

III. METHODS

A. Datasets and Labels

HAM10000 [6] is used for training/validation and DDI [4], [5] as a held-out test set — a deliberate train/test domain shift (dermoscopic → clinical photography). Labels are binarized to **Benign** vs **Malignant** (HAM10000 `dx == mel` → Malignant; DDI `malignant` field). DDI Fitzpatrick codes are mapped to

three groups: Light (FST I–II), Medium (FST III–IV), and Dark (FST V–VI). Table I summarizes the splits.

TABLE I. Dataset Summary

Dataset	Total	Malignant	Benign	Skin tone
HAM10000 (train)	8012	884	7128	—
HAM10000 (val)	2003	229	1774	—
DDI (Light)	208	49	159	FST I–II
DDI (Medium)	241	74	167	FST III–IV
DDI (Dark)	207	48	159	FST V–VI

Images are converted to RGB, resized to 224×224, and normalized with ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]). The HAM10000 train/validation split is a stratified 80/20 split performed **by `lesion_id`** so that no lesion appears in both splits (preventing leakage), with seed 42.

B. Model and Training

The classifier is EfficientNet-B0 initialized with ImageNet weights, with its final layer replaced by a two-class head, `Linear(1280 → 2)` (Fig. 1). Only the head is new; the backbone is fine-tuned. Training uses class-weighted cross-entropy (inverse-frequency weights to counter benign-heavy imbalance), the Adam optimizer (learning rate 1×10^{-4}), and a `ReduceLROnPlateau` scheduler on validation F1. The best checkpoint is selected by validation F1, and training early-stops when validation F1 stops improving.

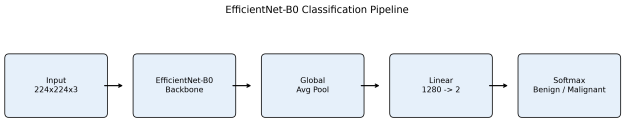


Fig. 1. End-to-end pipeline: image preprocessing, EfficientNet-B0 backbone, two-class head, and the Grad-CAM branch on the final convolutional block.

C. Skin-Tone-Aware Augmentation

Augmentation is applied **to the training set only** in the augmented experiment, before normalization, in random order: `RandomHorizontalFlip(0.5)`, `RandomRotation(±20°)`, and `ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2, hue=0.1)`. The color operations sample factors from uniform ranges and transform the image as follows, where `in`/`out` are pixel values, $\bar{g}$ is mean luminance, and `gray` is the grayscale image:

$$\textbf{brightness: out} = b \cdot \textbf{in}, \quad b \sim U(0.8, 1.2) \tag{1}$$

$$\textbf{contrast: out} = c\,(\textbf{in} - \bar{g}) + \bar{g}, \quad c \sim U(0.8, 1.2) \tag{2}$$

$$\textbf{saturation: out} = s\,\textbf{in} + (1 - s)\,\textbf{gray}, \quad s \sim U(0.8, 1.2) \tag{3}$$

$$\textbf{hue: shift } H \textbf{ in HSV by } h \sim U(-0.1, 0.1)\ (\approx \pm 36°) \tag{4}$$

Darkening arises specifically from a brightness factor $b < 1$ (and, secondarily, reduced contrast/saturation), approximating darker-skin appearance in the absence of skin-tone labels in HAM10000. Representative outputs are shown in Fig. 2.

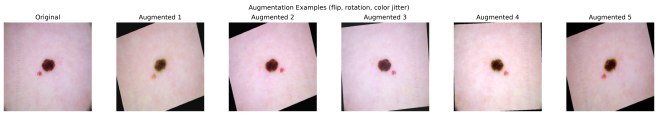


Fig. 2. One training lesion rendered under several augmentations (flip, rotation, color jitter).

D. Grad-CAM Explainability

Grad-CAM is computed on the last convolutional block (`model.features[-1]`). For a target class c with activation maps A^k (channel k), the gradient of the class score y^c is global-average-pooled to channel weights α_k^c, combined, and rectified:

$$\alpha_k^c = \frac{1}{Z}\sum_i\sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{5}$$

$$L^c = \mathrm{ReLU}\Big(\sum_k \alpha_k^c A^k\Big) \tag{6}$$

L^c is bilinearly upsampled to 224×224, min–max normalized to $[0, 1]$, and alpha-blended over the input as $(1-\alpha)\,\textbf{image} + \alpha\,\textbf{jet}(L^c)$ with $\alpha = 0.4$.

IV. EXPERIMENTAL SETUP

Two experiments are run with identical evaluation: **baseline** (preprocessing-only training) and **augmented** (Section III-C). Both are trained on HAM10000 and evaluated on all 656 DDI images. Hyperparameters: batch size 32, Adam learning rate 1×10^{-4}, seed 42, up to 15 epochs with early stopping on validation F1. With batch size 32, one epoch is 251 training batches (8012 images) and 63 validation batches (2003 images). Metrics are Accuracy, Precision, Recall, F1, and ROC-AUC (positive class = Malignant), plus per-skin-tone accuracy and a Light–Dark fairness gap. Training was CPU-bound; the baseline early-stopped near epoch 6 (best validation F1 at epoch 3). All artifacts (checkpoints, metrics JSON, figures, tables) are regenerated deterministically from the pipeline.

V. RESULTS

A. Overall Performance

On the DDI test set, augmentation substantially improved malignant detection. The baseline is extremely conservative — perfect precision but near-zero recall — whereas the augmented model trades some precision for much higher recall, F1, and ROC-AUC (Table II). ROC curves and confusion matrices are shown in Fig. 4 and Fig. 5; training curves in Fig. 3.

TABLE II. Overall DDI Metrics (Baseline vs Augmented)

Metric	Baseline	Augmented
Accuracy	0.745	0.741
Precision	1.000	0.512
Recall	0.023	0.123
F1	0.046	0.198
ROC-AUC	0.554	0.615

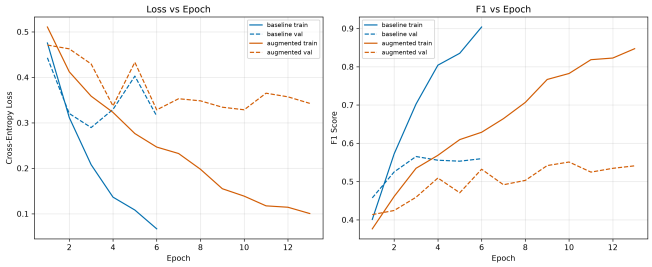


Fig. 3. Training/validation loss and F1 across epochs.

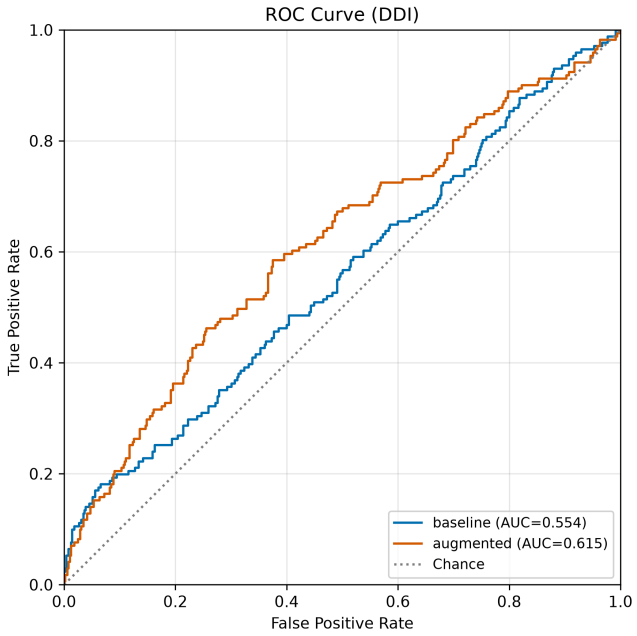


Fig. 4. ROC curves (DDI) for the baseline and augmented models.

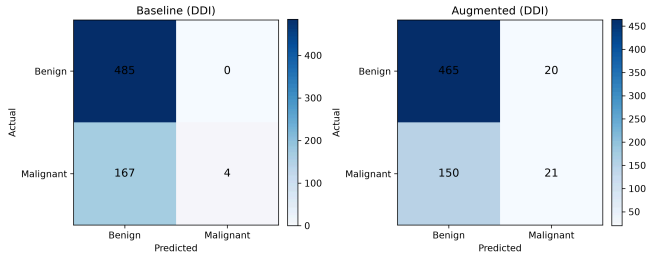


Fig. 5. Confusion matrices (DDI) for the baseline and augmented models.

B. Fairness Across Skin Tones

Per-group results are reported in Table III, and the fairness gaps in Table IV. Augmentation raised Light and Medium accuracy but **lowered** Dark accuracy (0.778 → 0.725). As a result, the overall max–min spread narrowed (0.077 → 0.049), but the primary Light–Dark gap did not improve and in fact became positive (−0.013 → +0.049). Per-group recall/F1 (Table III) did rise for every group, consistent with the overall sensitivity gain.

TABLE III. Per-Skin-Tone Metrics

Model	Skin tone	Count	Accuracy	Precision	Recall	F1
Baseline	Light	208	0.764	0.000	0.000	0.000
Baseline	Medium	241	0.701	1.000	0.027	0.053
Baseline	Dark	207	0.778	1.000	0.042	0.080
Augmented	Light	208	0.774	0.750	0.061	0.113
Augmented	Medium	241	0.726	0.722	0.176	0.283
Augmented	Dark	207	0.725	0.263	0.104	0.149

TABLE IV. Fairness Gap (Accuracy by Group)

Model	Light	Medium	Dark	Light−Dark	Max−Min
Baseline	0.764	0.701	0.778	−0.013	0.077
Augmented	0.774	0.726	0.725	+0.049	0.049

C. Explainability

Grad-CAM overlays (Fig. 6) across skin tones and both classes confirm the model generally concentrates activation on the lesion region rather than surrounding skin or imaging artifacts, supporting that predictions are lesion-driven.

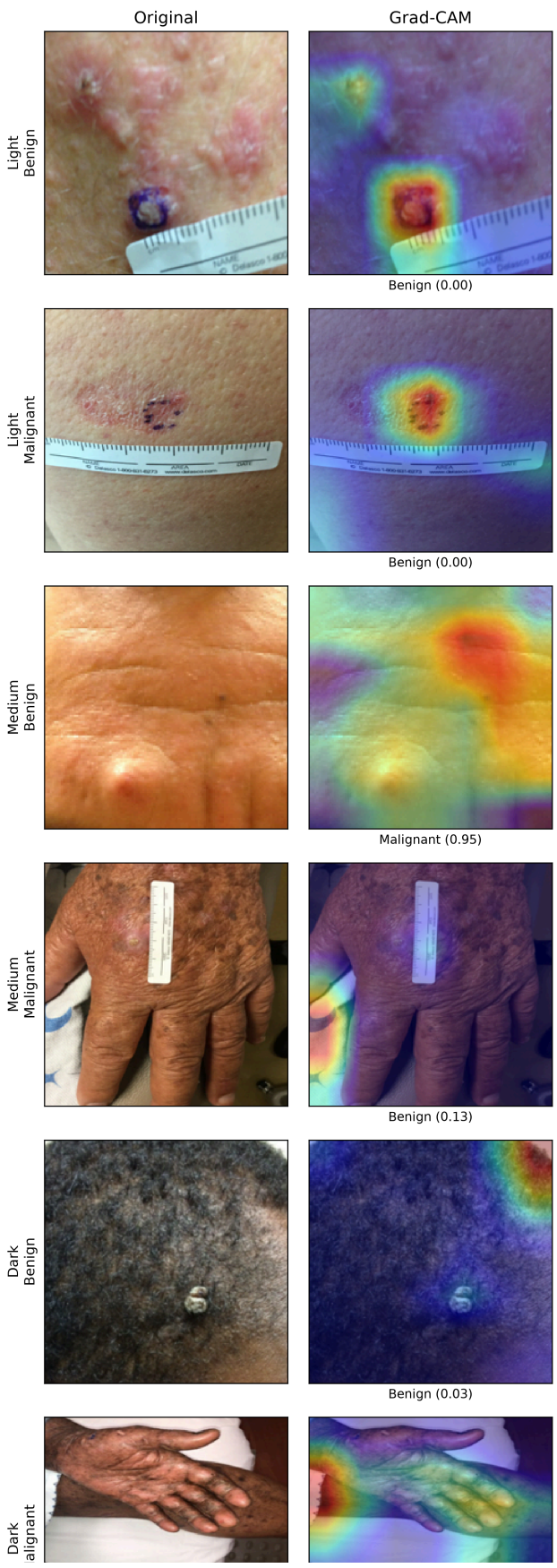


Fig. 6. Grad-CAM examples across skin tones (Light/Medium/Dark) and classes (Benign/Malignant).

VI. DISCUSSION

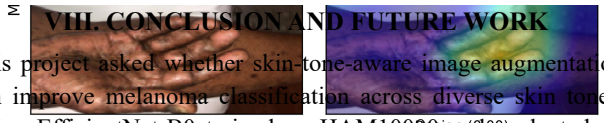
Augmentation clearly improved the model's ability to *detect* malignancy: recall rose from 0.023 to 0.123, F1 from 0.046 to 0.198, and ROC-AUC from 0.554 to 0.615. The baseline behaved as a near-constant "benign" predictor (perfect precision, negligible recall), which inflates accuracy on a benign-heavy test set while being clinically useless; augmentation produced a more balanced operating point.

The fairness picture is mixed. The max–min accuracy spread narrowed, but the targeted Light–Dark accuracy gap did **not** improve — Dark-group accuracy actually fell even though Dark-group recall and F1 increased. This apparent tension is a consequence of the benign-heavy class balance: accuracy and sensitivity move differently as the decision threshold shifts. The most plausible explanation is that photometric jitter changes pixel intensities but not lesion morphology or the characteristics of real skin-of-color clinical photography, so it is a weak proxy for genuine diversity.

This aligns with the DDI study [4], where the large fairness gains came specifically from **fine-tuning on diverse DDI data** (HAM10000 Light/Dark ROC-AUC rising to ≈0.77/0.78), not from color augmentation of a non-diverse corpus. Our augmented model never trains on DDI, so it should not be expected to reproduce those gains. Our baseline DDI ROC-AUC (0.554) is also lower than the study's HAM10000 baseline (0.67), attributable to a different architecture (EfficientNet-B0 vs Inception V3), a single seed, class-weighted loss versus balanced sampling, early stopping, and different preprocessing/decision thresholds.

VII. LIMITATIONS

- **Different source datasets.** HAM10000 (train) and DDI (test) differ in acquisition, population, and label distribution; cross-dataset shift affects absolute metrics.
- **No skin-tone labels in HAM10000.** Fairness cannot be optimized during training, only evaluated on DDI.
- **Photometric augmentation ≠ real SOC data.** Color/brightness jitter approximates but does not replicate real skin-of-color imagery. The ±0.2 range only darkens up to ~20%, a weak proxy for true dark-skin appearance.
- **DDI is small.** Limited per-group counts (≈207–241) yield high-variance fairness estimates.
- **Grad-CAM is localization, not segmentation.** Heatmaps indicate influential regions, not precise lesion boundaries, and are not a diagnostic segmentation.
- **Research-only.** This is not a medical device; outputs must not be used for clinical diagnosis. Results also reflect a single seed and CPU-limited training.



VIII. CONCLUSION AND FUTURE WORK

This project asked whether skin-tone-aware image augmentation can improve melanoma classification across diverse skin tones. Using EfficientNet-B0 trained on HAM10000 and evaluated on the skin-tone-labeled DDI benchmark, augmentation improved overall malignant detection (F1 and ROC-AUC) and narrowed the max–min per-group accuracy spread, but did **not** reduce the targeted Light–Dark gap. The hypothesis is therefore only partially supported: photometric augmentation helps robustness but is insufficient, on its own, to close skin-tone fairness gaps — consistent with prior findings that representative data is the key lever [4].

Future work: fine-tune on genuinely diverse data (as in [4]) or synthesize dark-skin imagery; apply stronger/targeted augmentation with Mixup; run multiple seeds with confidence intervals; train on the full dataset with GPU acceleration; and explore fairness-aware objectives (e.g., GroupDRO, CORAL). A promising direction is adopting datasets that provide pixel-level lesion masks (e.g., ISIC/PH2) so that **segmentation** can replace Grad-CAM for more precise lesion localization and potentially more robust classification.

ACKNOWLEDGMENT

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